AP, GP AND AGP

(Arithmetico-Geometric Progression)

JEE Advanced Preparation — Mathematics

AP properties | GP properties | AGP derivation | Infinite sums | Means insertion | Special sums

Subject	Mathematics
Level	JEE Advanced
Topic	AP, GP and Arithmetico-Geometric Progression (AGP)
Key Methods	Sum formulae; Means insertion; AGP multiply-subtract trick

1. ARITHMETIC PROGRESSION (AP)

Definition: A sequence a, a+d, a+2d, ... where d = common difference.

nth term: $T_n = a + (n-1)d$

Sum of n terms: $S_n = n/2 * [2a + (n-1)d] = n/2 * [T_1 + T_n]$

Important properties:

1. $T_m + T_n = T_p + T_q$ whenever $m+n = p+q$
2. If a, b, c are in AP then $2b = a+c$ (b is AM of a and c)
3. Three numbers in AP: take them as (a-d), a, (a+d) [sum = 3a]
4. Four numbers in AP: (a-3d), (a-d), (a+d), (a+3d) [sum = 4a]
5. Sum of n terms: $S_n = An^2 + Bn$ for some constants A, B

=> $T_n = S_n - S_{n-1} = A(2n-1) + B$ (linear in n)

1.1 Inserting n Arithmetic Means Between a and b

Insert n AMs $A_1, A_2, ..., A_n$ between a and b.

The full sequence a, $A_1, A_2, ..., A_n$, b is an AP of (n+2) terms.

Common difference: $D = (b-a)/(n+1)$

$A_k = a + k*D = a + k*(b-a)/(n+1)$

Sum of all n AMs: Sum = $n*(a+b)/2$ [AM of a and b, times n]

Note: Sum of ALL n+2 terms = $(n+2)*(a+b)/2$

2. GEOMETRIC PROGRESSION (GP)

Definition: A sequence a, ar, ar^2, ... where r = common ratio.

nth term: $T_n = a*r^{(n-1)}$

Sum of n terms:

$S_n = a*(r^n - 1)/(r-1)$ for $r \neq 1$

$S_n = na$ for $r = 1$

Sum to infinity: $S_{inf} = a/(1-r)$ for $|r| < 1$

Important properties:

1. $T_m * T_n = T_p * T_q$ whenever $m+n = p+q$
2. If a, b, c are in GP then $b^2 = ac$ (b is GM of a and c)
3. Three numbers in GP: take them as a/r, a, a*r [product = a^3]
4. Product of n terms = $(T_1 * T_n)^{(n/2)}$ = $(first * last)^{(n/2)}$
5. Log of GP terms form an AP.

2.1 Inserting n Geometric Means Between a and b

Insert n GMs $G_1, G_2, \ldots, G_n$ between a and b ($a, b > 0$).

The full sequence $a, G_1, \ldots, G_n, b$ is a GP of $(n+2)$ terms.

Common ratio: $R = (b/a)^{1/(n+1)}$

$G_k = a * R^k = a * (b/a)^{k/(n+1)}$

Product of all n GMs = $(ab)^{n/2}$

Product of all n+2 terms = $(ab)^{(n+2)/2}$

3. AP vs GP — KEY COMPARISONS

Property	AP	GP
nth term	$a+(n-1)d$	$a*r^{n-1}$
Sum of n terms	$n/2*[2a+(n-1)d]$	$a*(r^n-1)/(r-1)$
Sum to infinity	Diverges	$a/(1-r)$ if $\|r\|<1$
Condition for 3 terms	$2b=a+c$	$b^2=ac$
Take 3 terms as	a-d, a, a+d	a/r, a, a*r
Consecutive mean	AM	GM
Means inserted	$D=(b-a)/(n+1)$	$R=(b/a)^{1/(n+1)}$
Common difference/ratio	Constant difference	Constant ratio

4. ARITHMETICO-GEOMETRIC PROGRESSION (AGP)

DEFINITION: An AGP is formed by multiplying corresponding terms of an AP and GP.

AP: $a, a+d, a+2d, \ldots, a+(n-1)d$

GP: $1, r, r^2, \ldots, r^{(n-1)}$

AGP: $a, (a+d)r, (a+2d)r^2, \ldots, (a+(n-1)d) \cdot r^{(n-1)}$

General term: $T_k = (a+(k-1)d) \cdot r^{(k-1)}$

4.1 Sum of n Terms of AGP — Derivation

The Multiply-and-Subtract Trick:

Let $S_n = a + (a+d)r + (a+2d)r^2 + \ldots + (a+(n-1)d)r^{(n-1)}$ ...(1)

Multiply both sides by r:

$r \cdot S_n = a \cdot r + (a+d)r^2 + \ldots + (a+(n-2)d)r^{(n-1)} + (a+(n-1)d)r^n$ ...(2)

Subtract (2) from (1): $S_n - r \cdot S_n = S_n \cdot (1-r)$

$= a + d \cdot r + d \cdot r^2 + \ldots + d \cdot r^{(n-1)} - (a+(n-1)d) \cdot r^n$

$= a + d \cdot r \cdot (1-r^{(n-1)})/(1-r) - (a+(n-1)d) \cdot r^n$ [GP sum for d terms]

Therefore ($r \neq 1$):

$S_n = a/(1-r) + d \cdot r \cdot (1-r^{(n-1)})/(1-r)^2 - (a+(n-1)d) \cdot r^n/(1-r)$

4.2 Sum to Infinity of AGP (|r| < 1)

As $n \to \infty$, $r^n \to 0$ and $r^{(n-1)} \to 0$ (since $|r| < 1$):

$S_{inf} = a/(1-r) + d \cdot r/(1-r)^2$

This is a very important formula for JEE!

Special case: $1+2x+3x^2+4x^3+\ldots$ for $|x|<1$

This is an AGP with $a=1$, $d=1$, $r=x$:

$S_{inf} = 1/(1-x) + x/(1-x)^2 = [(1-x)+x]/(1-x)^2 = 1/(1-x)^2$

Sum = $1/(1-x)^2$ for $|x| < 1$

Also derived by differentiating $\mathrm{Sum}(x^n) = 1/(1-x)$: d/dx gives $\mathrm{Sum}\ n \cdot x^{(n-1)} = 1/(1-x)^2$

4.3 Standard Special AGP: Sum r/2^r

$S = \mathrm{Sum}(r=1 \text{ to } n)\ r/2^r = 1/2 + 2/4 + 3/8 + \ldots + n/2^n$

This is AGP with $a=1$, $d=1$, $r=1/2$.

$S = a/(1-r) + d \cdot r \cdot (1-r^{(n-1)})/(1-r)^2 - (a+(n-1)d) \cdot r^n/(1-r)$

$= 1/(1/2) + (1/2) \cdot (1-(1/2)^{(n-1)})/(1/4) - n \cdot (1/2)^n/(1/2)$

$= 2 + 2*(1-(1/2)^{(n-1)}) - 2n/2^n$

$= 2 + 2 - 2/2^{(n-1)} - 2n/2^n$

$= 4 - 4/2^n - 2n/2^n = 4 - (2n+4)/2^n$

$S_n = 2 - (n+2)/2^{(n-1)}$ [or equivalently $4-(2n+4)/2^n$]

$S_{inf} = 2$ (as $n \to inf$)

5. WORKED EXAMPLES

Example 1: Find sum to n terms and to infinity: $1 + 2*(1/2) + 3*(1/4) + 4*(1/8) + ...$

Solution:

This is AGP with $a=1$, $d=1$, $r=1/2$

Use multiply-subtract method:

$S = 1 + 2*(1/2) + 3*(1/4) + ... + n*(1/2)^{(n-1)}$

$(1/2)*S = 1*(1/2) + 2*(1/4) + ... + (n-1)*(1/2)^{(n-1)} + n*(1/2)^n$

$S - S/2 = 1 + (1/2) + (1/4) + ... + (1/2)^{(n-1)} - n/2^n$

$S/2 = [1*(1-(1/2)^n)/(1-1/2)] - n/2^n = 2*(1-1/2^n) - n/2^n$

$S/2 = 2 - 2/2^n - n/2^n = 2 - (n+2)/2^n$

ANSWER: $S_n = 4 - (n+2)/2^{(n-1)}$; $S_{inf} = 4$ (as n->inf, second term->0)

Wait: $S/2 = 2-(n+2)/2^n \Rightarrow S = 4 - (n+2)/2^{(n-1)}$. $S_{inf} = 4$.

Example 2: Find sum to n terms: $1 + 3x + 5x^2 + 7x^3 + ...$ (assume $|x| < 1$ for infinity)

Solution:

AP part: 1, 3, 5, 7, ... so $a=1$, $d=2$

GP part: $1, x, x^2, x^3, ...$ so $r=x$

$S_n = a + (a+d)x + (a+2d)x^2 + ... + (a+(n-1)d)x^{(n-1)}$

Multiply by x: $x*S_n = ax + (a+d)x^2 + ... + (a+(n-2)d)x^{(n-1)} + (a+(n-1)d)x^n$

$S_n - x*S_n = a + d*x + d*x^2 + ... + d*x^{(n-1)} - (a+(n-1)d)x^n$

$S_n*(1-x) = 1 + 2x*(1-x^{(n-1)})/(1-x) - (2n-1)*x^n$

ANSWER (infinite sum $|x|<1$): $S_{inf} = 1/(1-x) + 2x/(1-x)^2$

$= [(1-x) + 2x]/(1-x)^2 = (1+x)/(1-x)^2$

Example 3: Insert 4 arithmetic means between 3 and 23. Find their sum.

Solution:

Sequence: $3, A_1, A_2, A_3, A_4, 23$ — total 6 terms in AP

Common difference $D = (23-3)/(4+1) = 20/5 = 4$

$A_1 = 3+4 = 7$

$A_2 = 7+4 = 11$

$A_3 = 11+4 = 15$

$A_4 = 15+4 = 19$

Sum of AMs = 7+11+15+19 = 52

OR: Sum $= n*(a+b)/2 = 4*(3+23)/2 = 4*13 = 52$ [confirmed!]

ANSWER: AMs are 7, 11, 15, 19 and their sum is 52

Example 4: Insert 3 geometric means between 2 and 162. Find product of all 5 terms.

Solution:

Sequence: 2, G_1, G_2, G_3, 162 — total 5 terms in GP

Common ratio R = (162/2)^(1/(3+1)) = 81^(1/4) = (3^4)^(1/4) = 3

G_1 = 2*3 = 6

G_2 = 6*3 = 18

G_3 = 18*3 = 54

Check: 54*3 = 162 [confirmed!]

Product of all 5 terms = 2*6*18*54*162

OR: Product = (first * last)^(n/2) = (2*162)^(5/2) = 324^(5/2) = (18^2)^(5/2) = 18^5

18^5 = 18*18*18*18*18 = 1889568

ANSWER: GMs are 6, 18, 54; product of all 5 terms = 18^5 = 1889568

Example 5: Find sum: 1*2^0 + 2*2^1 + 3*2^2 + ... + n*2^(n-1)

Solution:

AGP with a=1, d=1, r=2

Let S = 1 + 2*2 + 3*4 + 4*8 + ... + n*2^(n-1)

2*S = 1*2 + 2*4 + 3*8 + ... + (n-1)*2^(n-1) + n*2^n

S - 2*S = 1 + 1*2 + 1*4 + ... + 1*2^(n-1) - n*2^n

-S = (2^n - 1)/(2-1) - n*2^n [GP sum = 2^n - 1]

-S = 2^n - 1 - n*2^n = (1-n)*2^n - 1

S = (n-1)*2^n + 1

ANSWER: S_n = (n-1)*2^n + 1

Check n=3: 1+4+12=17; formula: 2*8+1=17 [confirmed!]

Example 6: If a,b,c are in AP and a,b,d are in GP (d =/= b), find c in terms of a and b

Solution:

a, b, c in AP => b - a = c - b => c = 2b - a

a, b, d in GP => b^2 = a*d => d = b^2/a

So c = 2b - a

Additional insight: relation between c and d:

c - b = b - a (AP condition)

d/b = b/a (GP condition) => d = b^2/a

ANSWER: c = 2b - a

The value of d = b^2/a (from GP condition)

Note: If a=1, b=2: c=3, d=4; AP: 1,2,3; GP: 1,2,4 [classic example]

6. PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. [Single Correct] The sum to infinity of 1 + 2*(1/3) + 3*(1/9) + 4*(1/27) + ... is:

(A) 3/2
(B) 9/4
(C) 2
(D) 3

Q2. [Single Correct] Sum to n terms of 1*2^0 + 2*2^1 + 3*2^2 + ... + n*2^(n-1) is:

(A) (n+1)*2^n - 1
(B) (n-1)*2^n + 1
(C) n*2^n
(D) (n-1)*2^(n+1)

Q3. [Single Correct] 4 arithmetic means are inserted between 3 and 23. The sum of AMs is:

(A) 48
(B) 52
(C) 56
(D) 44

Q4. [Single Correct] If three numbers are in GP with product 216, the middle number is:

(A) 4
(B) 6
(C) 9
(D) 8

Q5. [Multiple Correct] If a,b,c are in AP then:

(A) 2b = a+c
(B) b = (a+c)/2
(C) b^2 = ac
(D) a,b,c can be consecutive integers

Q6. [Single Correct] S_inf of the AGP 1+2x+3x^2+4x^3+... for |x|<1 is:

(A) 1/(1-x)
(B) 1/(1-x)^2
(C) x/(1-x)^2
(D) (1+x)/(1-x)^2

Q7. [Single Correct] Sum(r=1 to inf) r/2^r equals:

(A) 1
(B) 2
(C) 4
(D) 1/2

Q8. [Single Correct] 3 geometric means are inserted between 2 and 162. The third GM is:

(A) 18
(B) 54
(C) 6
(D) 486

Q9. [Multiple Correct] For an AP with first term a and common difference d:

(A) T_n = a+(n-1)d

(B) Sum $S_n = n/2*(T_1+T_n)$

(C) $T_m+T_n=T_{m+n}$ always

(D) Sum of n AMs between a,b = $n*(a+b)/2$

Q10. [Single Correct] Sum to n terms of $1+3x+5x^2+7x^3+...$ ($|x|<1$) to infinity is:

(A) $1/(1-x)$

(B) $(1+x)/(1-x)^2$

(C) $2/(1-x)^2$

(D) $(1-x)/(1+x)^2$

7. ANSWER KEY WITH SOLUTIONS

Q#	Answer	Key Solution
Q1	(B) 9/4	AGP inf: a/(1-r)+dr/(1-r)^2 = 1/(2/3)+(1/3)/(4/9) = 3/2+3/4=9/4
Q2	(B) (n-1)*2^n+1	Multiply-subtract: S=2+(n-1)*2^n where we showed S=(n-1)*2^n+1
Q3	(B) 52	n*(a+b)/2 = 4*(3+23)/2 = 4*13 = 52
Q4	(B) 6	Take GP as a/r, a, a*r; product = a^3 = 216; a = 6 (middle term)
Q5	A, B, D	(C) b^2=ac is GP condition, not AP
Q6	(B) 1/(1-x)^2	AGP: 1/(1-x)+x/(1-x)^2 = 1/(1-x)^2
Q7	(B) 2	S_inf of AGP with a=1,d=1,r=1/2: 1/(1/2)+(1/2)/(1/4)=2+2=4? Actually S_n=(n+2)/2^(n-1) subtracted from 4... S_inf=2. a/(1
Q8	(B) 54	R=3; G1=6, G2=18, G3=54
Q9	A, B, D	(C) T_{m+n}=a+(m+n-1)d =/= T_m+T_n in general
Q10	(B) (1+x)/(1-x)^2	AGP a=1,d=2,r=x: 1/(1-x)+2x/(1-x)^2=(1-x+2x)/(1-x)^2=(1+x)/(1-x)^2

8. KEY FORMULAS — QUICK REFERENCE

Concept	Formula
AP nth term	T_n = a + (n-1)d
AP sum n terms	S_n = n/2 * [2a + (n-1)d] = n/2*(T_1+T_n)
n AMs between a,b	D=(b-a)/(n+1); sum of AMs = n*(a+b)/2
GP nth term	T_n = a*r^(n-1)
GP sum n terms	S_n = a*(r^n-1)/(r-1)
GP sum to infinity	S_inf = a/(1-r) for |r|<1
n GMs between a,b	R=(b/a)^(1/(n+1)); product of GMs = (ab)^(n/2)
AGP general term	T_k = (a+(k-1)d)*r^(k-1)
AGP sum to infinity	S_inf = a/(1-r) + dr/(1-r)^2 for |r|<1
1+2x+3x^2+...	= 1/(1-x)^2 for |x|<1
Sum r/2^r (r=1 to n)	4 - (2n+4)/2^n = 2 - (n+2)/2^(n-1)
Sum r/2^r (r=1 to inf)	= 2
1+3x+5x^2+... inf	= (1+x)/(1-x)^2 for |x|<1

ADVANCED NOTES: AP, GP and AGP

AP-GP-HP Interrelations

If a, b, c are in AP: b = (a+c)/2 => a+c = 2b

If a, b, c are in GP: b^2 = ac

If a, b, c are in HP: b = 2ac/(a+c)

Key identity: If a, b, c in GP, then a+c >= 2b (AM >= GM for a,c).

Combining: a, A, b in AP and a, G, b in GP => A >= G (AM >= GM).

For positive a,b: AM*HM = GM^2 (i.e., (a+b)/2 * 2ab/(a+b) = ab = GM^2).

Recurring Decimals as Infinite GPs

0.333... = 3/10 + 3/100 + 3/1000 + ... = (3/10)/(1-1/10) = 1/3.

0.142857142857... (1/7) = infinite GP identification.

General recurring decimal: 0.abcabc... = abc/999.

Non-pure recurring: 0.1333... = (13-1)/90 = 12/90 = 2/15.

JEE occasionally tests conversion of recurring decimals to fractions using GP sum.

JEE ADVANCED EXAM STRATEGY

Recognise the Type First: Before computing, classify whether it is AP/GP/AGP/Special series.

Use Standard Results: Memorise S1, S2, S3, and the product series formulas — saves huge time.

Telescoping Check: If T_r has product of consecutive integers, try partial fractions first.

AM-GM Shortcut: For min/max with a constraint, always try AM-GM before calculus.

Verification: Always verify with n=1 or n=2 before writing final answer.

AGP Infinite Sum: If |r|<1, use S_inf = a/(1-r) + dr/(1-r)^2 directly.

Nicomachus: Any time S3 appears, immediately replace with (S1)^2.

Multiple Correct MCQs: All four options may be true — verify each independently.

COMPLETE FORMULA CONSOLIDATION

#	Series / Expression	Closed Form
1	Sum r (r=1 to n)	n(n+1)/2
2	Sum r^2 (r=1 to n)	n(n+1)(2n+1)/6
3	Sum r^3 (r=1 to n)	[n(n+1)/2]^2
4	Sum r(r+1)	n(n+1)(n+2)/3

5	Sum r(r+1)(r+2)	n(n+1)(n+2)(n+3)/4
6	Sum r*r! (r=1 to n)	(n+1)! - 1
7	Sum odd numbers (n terms)	n^2
8	Sum (2r-1)^2 (r=1 to n)	n(2n-1)(2n+1)/3
9	Sum (2r)^2 (r=1 to n)	2n(n+1)(2n+1)/3
10	Sum (2r-1)^3 (r=1 to n)	n^2(2n^2-1)
11	1/(r(r+1)) — telescoping	Sum to n = n/(n+1)
12	r/(r+1)! — telescoping	Sum to n = 1-1/(n+1)!
13	AP nth term	a+(n-1)d
14	AP sum n terms	n[2a+(n-1)d]/2
15	GP nth term	a*r^(n-1)
16	GP sum infinity	a/(1-r) for $\vert r\vert<1$
17	AGP sum infinity	a/(1-r) + dr/(1-r)^2
18	1+2x+3x^2+... ($\vert x\vert<1$)	1/(1-x)^2
19	Sum r/2^r (r=1 to inf)	2
20	AM >= GM >= HM	Always; equality iff all equal